



CESIM TEACHING NOTE

Capital Budgeting Microsimulation

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1.Simulation Overview

The **Capital Budgeting Simulation** provides an engaging, hands-on experience in strategic financial decision-making. Participants step into the role of senior executives tasked with optimizing investment portfolios across diverse business areas, applying key tools like NPV, IRR, and risk-adjusted discount rates.

Structured as a single-player microsimulation, it challenges users to assess a variety of projects under budget constraints, weighing trade-offs between risk and return, and short-term outcomes versus long-term value creation. The experience builds critical skills in financial analysis, resource allocation, and strategic planning—within a realistic, evolving business context where every decision impacts overall organizational performance.

Learning Objectives

- Master capital budgeting fundamentals including NPV, IRR, payback period, and profitability index
- Develop risk assessment skills by evaluating project portfolios with varying risk profiles
- Apply discount rate estimation techniques based on project risk characteristics
- Practice strategic resource allocation under budget constraints
- Enhance critical thinking through trade-off analysis between competing investment opportunities
- Integrate theoretical knowledge with practical decision-making scenarios

Target Audience

- Undergraduate business students (accounting, finance, management)
- MBA students seeking practical accounting application
- Professional development programs for non-financial managers
- Prerequisites: Basic understanding of accounting principles and financial statement structure

Time Required

- 60-90 minutes for individual simulation completion
- 45-60 minutes for guided post-simulation debrief

Case Scenario and Winning Criteria

Participants engage in a simulated business environment where they evaluate and prioritize a portfolio of 15 strategic projects spanning areas such as Market Expansion, Operational Efficiency, Process Improvement, and Product Development. The simulation challenges users to make informed capital allocation decisions under budget constraints while navigating trade-offs between risk and return.

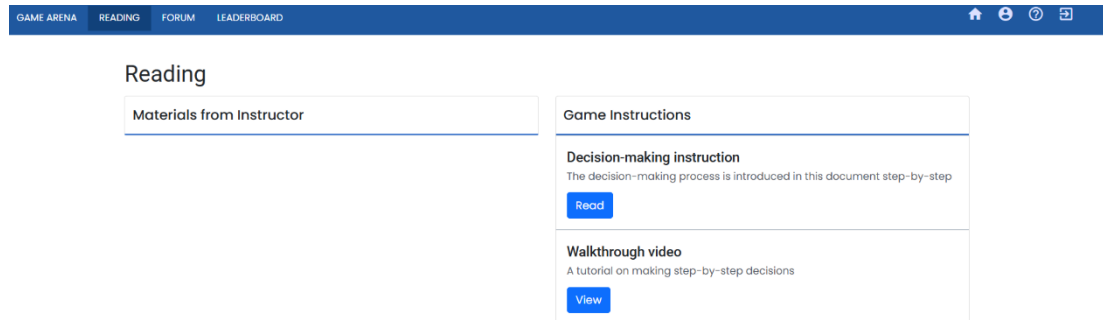
Winning Criteria (Based on three KPIs):

1. Net Present Value (NPV) of Cash Flows – Maximize overall value creation
2. Present Value (PV) of Cash Flows – Optimize total investment returns

3. Average Discount Rate Across Projects (%) – Achieve a balanced risk-return profile

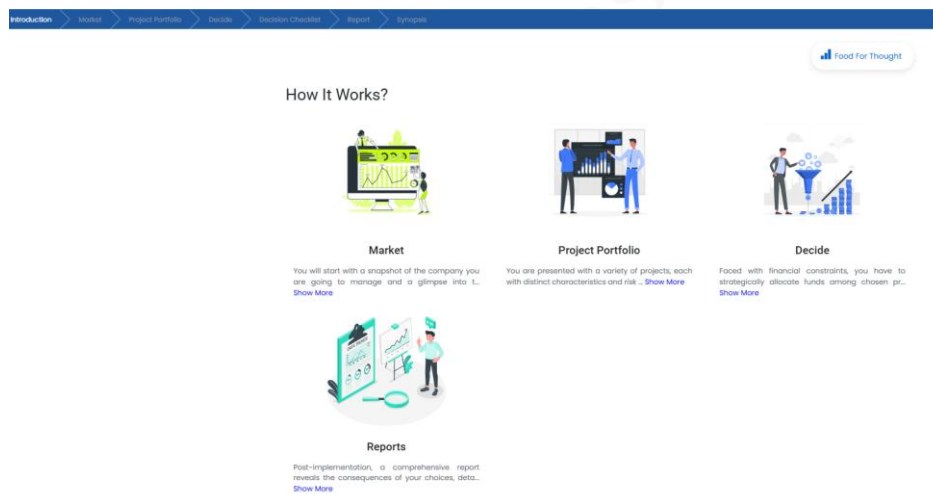
2. Pre- Simulation Preparation

a. Review Background Materials



- Login as participant to explore the platform
- Decision-making guide: Provides detailed explanations of each simulation section and decision point
- Walkthrough video: Offers visual demonstration of the simulation interface and decision process
- Industry overview: Understand the company context and business objectives

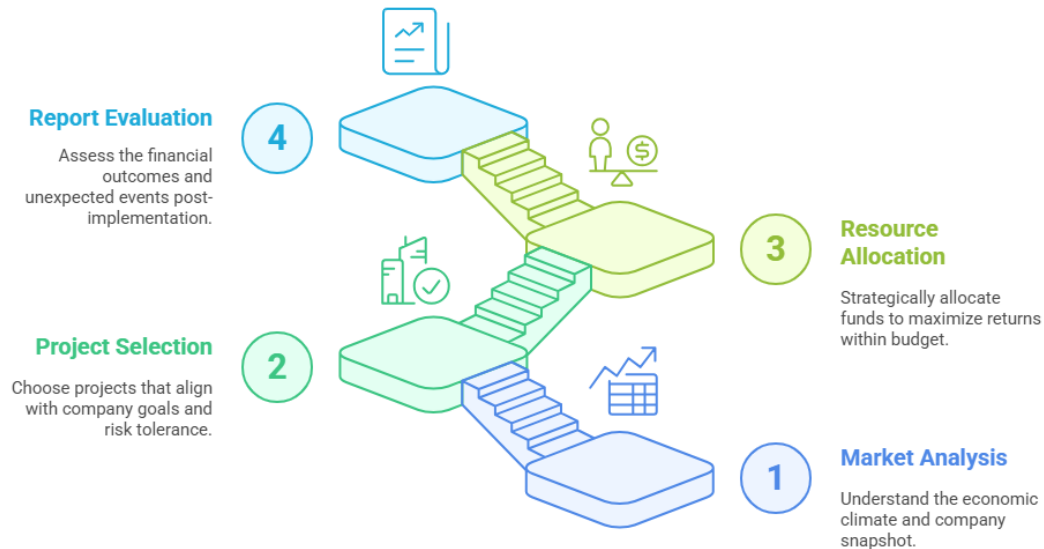
b. Familiarize with the Interface



- Game Arena: Central dashboard providing access to all decision areas
- Readings: Contains essential reference materials and guides
- Forum: Communication platform for participant questions and instructor feedback
- Decision tabs: Structured workflow from market analysis through financial decisions
- Reports: Performance analysis and feedback on previous decisions
- Synopsis: Summary of key decisions and outcomes

Understanding Decision-Making Process

The simulation follows a logical step-by-step process. Participants should move through each stage to make thoughtful decisions.



Core Algorithm: The simulation uses discounted cash flow analysis where:

- $NPV = \sum (Cash\ Flow / (1 + r)^t) - Initial\ Investment$
- Discount rates vary based on risk multipliers
- Budget constraint: Total selected projects $\leq$ INR 68 million

Key Instructor-Only Parameters:

- **Cash Flow Multipliers:** Risk-adjusted factors ranging from 0.91-0.99
- **Discount Rate Multipliers:** Risk premiums from 0.8-1.5
- **Market Parameters:** Fixed economic indicators (GDP growth 7.5%, inflation 4%, etc.)

Teaching with this Information:

- Emphasize how risk levels directly impact cash flow projections
- Demonstrate the relationship between risk tolerance and discount rate selection
- Show how market conditions influence project viability

Indicators (Available under the Market section of the Platform)

Macroeconomic Indicators	Value	Definition
Expected Repo Rate by RBI	5.5%	The interest rate at which the Reserve Bank of India lends money to commercial banks; serves as the benchmark for risk-free rate in discount rate calculations
Inflation Rate	4.0%	The annual percentage increase in general price levels; affects real returns and purchasing power of future cash flows

GDP Growth Rate	7.5%	The annual percentage increase in the country's economic output; indicates overall economic health and market expansion potential
Unemployment Rate	6.0%	The percentage of labor force that is unemployed; reflects economic conditions and potential impact on business operations
Exchange Rate (USD to INR)	75.5	The value of one US Dollar in Indian Rupees; affects international projects and foreign currency cash flows
Company-Specific Indicators		
Cost of Equity	13.0%	The minimum return required by equity investors; used to discount equity-financed projects and calculate WACC
Cost of Debt	9.0%	The effective interest rate the company pays on its borrowed funds; used for debt-financed projects and WACC calculation
Debt-to-Equity Ratio	40.0%	The proportion of debt financing relative to equity financing; indicates financial leverage and capital structure
Tax Rate	25.0%	The corporate income tax rate; affects after-tax cash flows and the tax shield benefit of debt financing
Weighted Average Cost of Capital	10.5%	The blended cost of all capital sources weighted by their proportions; serves as the baseline discount rate for average-risk projects
Budget (INR Million)	68	The total capital available for investment across all projects; represents the constraint within which portfolio optimization must occur

Calculations

- Discount rate = Hurdle rate (WACC)*Multiplier of the project based on risk; multiplier values are provided in instructor account >> click on card >> case management >> project portfolio
- Base WACC calculation: 10.5 % (derived from company's capital structure)
- Benefit-cost ratio = NPV/Initial Investment
- NPV = Present value of future cash flow - Initial Investment in present
- EAC = Initial Investment/Annuity Factor, where Annuity factor = $(1 - 1/(1 + \text{Discount rate})^{\text{period}})/\text{Discount rate}$.

Calculation on Base WACC calculation is as follows:

Formula for WACC = $(E/V * Re) + ((D/V * Rd * (1 - T_c))$ Where:

- E/V = Equity proportion (from Debt-to-Equity ratio)**
- D/ V = Debt proportion (from Debt-to-Equity ratio)**
- Re = Cost of equity** (return required by equity investors)
- Rd = Cost of debt** (interest rate on debt)
- Tc = Tax rate** (since interest on debt is tax-deductible)

Given **Debt-to-Equity ratio = 40%**, we assume:

- Equity = 100
- Debt = 40
- **Total Capital (V)** = Equity + Debt = 100 + 40 = 140

Therefore:

- $E/V = 100/140 = 0.714$ (Equity Proportion)
- $D/V = 40/140 = 0.286$ (Debt Proportion)
- Apply the WACC Formula
- $WACC = (E/V * Re) + ((D/V * Rd * (1 - Tc)))$
- $WACC = (0.714 \times 13\%) + (0.286 \times 9\% \times (1 - 25\%)) = 11.21$

By adjusting key financial factors, the base **WACC is considered to be 10.5% in the simulation**. Modifications in **market risk assumptions**, while slight deviations in the **debt-to-equity ratio** impacted capital weighting. Additionally, the influence of **macroeconomic conditions** such as the **repo rate (5.5%)** and **inflation (4.0%)**, along with internal risk assessments, resulted in a more conservative WACC estimate, ensuring alignment with market conditions and risk optimization strategies.

Risk Adjustment Multipliers

Risk Level	Discount Rate Multiplier	Cash Flow Multiplier
Low	0.9	1.0
Low-Moderate	1.02	0.97
Moderate	1.05	0.95
Moderate-High	1.07	0.93
High	1.09	0.91

Note: These multipliers are applied to base rates and expected cash flows to reflect project-specific risk characteristics. Higher risk projects receive higher discount rate multipliers (increasing the hurdle rate) and lower cash flow multipliers (reducing expected returns) to account for increased uncertainty and potential downside scenarios.

Common Participant Challenges:

- Overestimating returns by setting discount rates too low
- Ignoring portfolio risk diversification
- Focusing solely on NPV without considering risk balance
- Budget allocation inefficiencies

Instructor Pre-Implementation Checklist for Simulation Delivery

To ensure a smooth and impactful simulation experience for the learners, please review and complete the following preparatory steps:

- **Familiarize with the Simulation:** Complete a test run of the simulation to understand its structure, decision flow, and pacing. This allows to better anticipate participant questions and challenges.
- **Review Simulation Mechanics:** Examine the instructor-only backend settings and key variables (e.g., cost structures, demand drivers, elasticity assumptions) to understand how decisions influence outcomes.
- **Align with Course Objectives:** Identify the core concepts from the course that align with the simulation to reinforce key learning goals.
- **Prepare Contextual Examples:** Bring 2–3 real-world examples from relevant industries to illustrate successful and unsuccessful strategic decisions during the debrief.
- **Review Instructor Insights and Scoring:** Familiarize with the instructor account, leaderboard and performance indicators available to instructors. These tools will help guide feedback and encourage reflective learning.
- **Ensure Participant Access:** Confirm that all participants have registered using the right link / code and are able to access the simulation platform before the session begins.
- **Test Technical Readiness:** Coordinate with the IT team (if applicable) to ensure all devices, software, and internet connectivity are ready and compatible with the simulation requirements.

3. Facilitation Roadmap

Pre-Simulation Briefing

Case

Scenario:

Participants take on the role of senior executives at a growing insurance technology firm facing critical capital allocation decisions. With multiple projects across areas such as market expansion, operational efficiency, process improvement, and product innovation, each initiative carries distinct risks, investment needs, and strategic impact. Executives must evaluate and prioritize these opportunities within a constrained budget, aiming to build a balanced investment portfolio that supports long-term growth, risk diversification, and alignment with overall business objectives in a competitive landscape.

Introduction:

- Present the case scenario and company background
- Explain the simulation objectives focusing on capital budgeting analysis and strategic portfolio optimization
- Outline the decision-making process from market analysis through project evaluation to investment selection
- Clarify expectations regarding time commitment and submission deadlines

Contextual Guidance:

- Discuss the critical role of capital budgeting in corporate strategy and long-term value creation
- Highlight common investment challenges faced by technology companies in competitive markets
- Introduce the concept of portfolio optimization under budget constraints and strategic alignment
- Provide initial considerations for balancing quantitative analysis with strategic judgment

Technical Orientation:

- Demonstrate the simulation interface and sequential workflow
- Walk through the decision sequence from economic analysis to discount rate estimation to project selection
- Explain how to interpret project data, calculate financial metrics, and optimize budget allocation
- Address initial questions about navigation, calculation methods, risk assessment, and submission process

During Simulation**Monitoring Progress:**

- Track participant engagement across Market Analysis, Project Portfolio evaluation, and investment decisions
- Identify common challenges in discount rate estimation and portfolio optimization
- Provide hints for struggling participants without revealing optimal solutions
- Encourage systematic use of capital budgeting principles and verification of calculations

Intervention Strategies:

- Light intervention: Post general reminders about NPV calculations and risk assessment in the course forum
- Moderate intervention: Provide specific feedback on discount rate estimation approaches and budget utilization
- Strong intervention: Schedule individual consultation for participants facing significant analytical challenges

Mid-Simulation Guidance:

- Conduct brief check-in sessions to address common accounting questions
- Share anonymous examples of strong analytical approaches (without revealing answers)
- Highlight interesting portfolio allocation patterns observed in project selection
- Remind participants of budget constraints and the importance of strategic alignment review

Facilitation Interventions:

- "What factors can be considered in determining the discount rate for this high-risk project?"
- "How does the portfolio balance risk diversification with return maximization?"
- "What strategic rationale supports the selection of these particular projects?"
- "How do investment choices align with the company's growth objectives and risk tolerance?"

Debrief Structure

Performance Analysis:

- Review NPV maximization scores and portfolio optimization achievements across all participants
- Analyze the distribution of results and identify patterns in successful vs. suboptimal strategies
- Examine common errors in discount rate estimation and their impact on project rankings
- Present key performance metrics (NPV, PV, Average Discount Rate) and their relationships to decision quality

Strategic Discussion:

- Explore the rationale behind different approaches to risk assessment and discount rate determination
- Analyze investment portfolio patterns across Market Expansion, Operational Efficiency, Process Improvement, and Product Expansion categories
- Examine the interconnectedness of financial analysis, strategic planning, and resource allocation decisions

Learning Integration:

- Connect simulation experiences to fundamental capital budgeting principles and corporate finance theory
- Discuss real-world applications of portfolio optimization in corporate investment decisions
- Identify transferable insights for participants' future roles in financial analysis, strategic planning, and investment management
- Reflect on decision-making processes, analytical frameworks, and potential improvements in financial judgment under uncertainty

4. Critical Decision Points Analysis

Decision Point 1: Discount Rate Estimation

Key Decision: Determining appropriate discount rates for each project based on risk profiles

Optimal Approach:

Risk Level	Recommended Range	Rationale
Low	7.5-9.0%	Base rate + minimal risk premium
Low to Moderate	9.0-10.5%	Moderate risk adjustment

Moderate	10.5-12.0%	Standard risk premium
Moderate to High	12.0-13.5%	Higher risk compensation
High	13.5-15.0%	Maximum risk premium

Common Pitfalls:

- Using uniform discount rate across all projects
- Setting rates too low to inflate NPV artificially
- Ignoring project-specific risk characteristics

Decision Point 2: Risk-Return Portfolio Balance

Key Decision: Balancing high-return/high-risk projects with stable, lower-return options

Optimal Approach:

- Allocate 40-50% to moderate risk projects
- Include 20-30% high-return projects for growth
- Maintain 20-30% low-risk projects for stability

Portfolio Balance Table:

Project Category	Recommended Allocation	Risk Mitigation
Market Expansion	30-40%	Geographic diversification
Operational Efficiency	20-30%	Predictable returns
Process Improvement	20-25%	Cost reduction focus
Product Expansion	15-25%	Innovation premium

Common Pitfalls:

- Over-concentration in single category
- Ignoring portfolio diversification principles
- Chasing only high-NPV projects regardless of risk

Decision Point 3: Budget Allocation Strategy

Key Decision: Optimizing INR 68 million budget across 15 available projects

Optimal Approach:

- Prioritize projects with highest NPV per rupee invested
- Consider project synergies and strategic alignment
- Maintain budget buffer for risk management

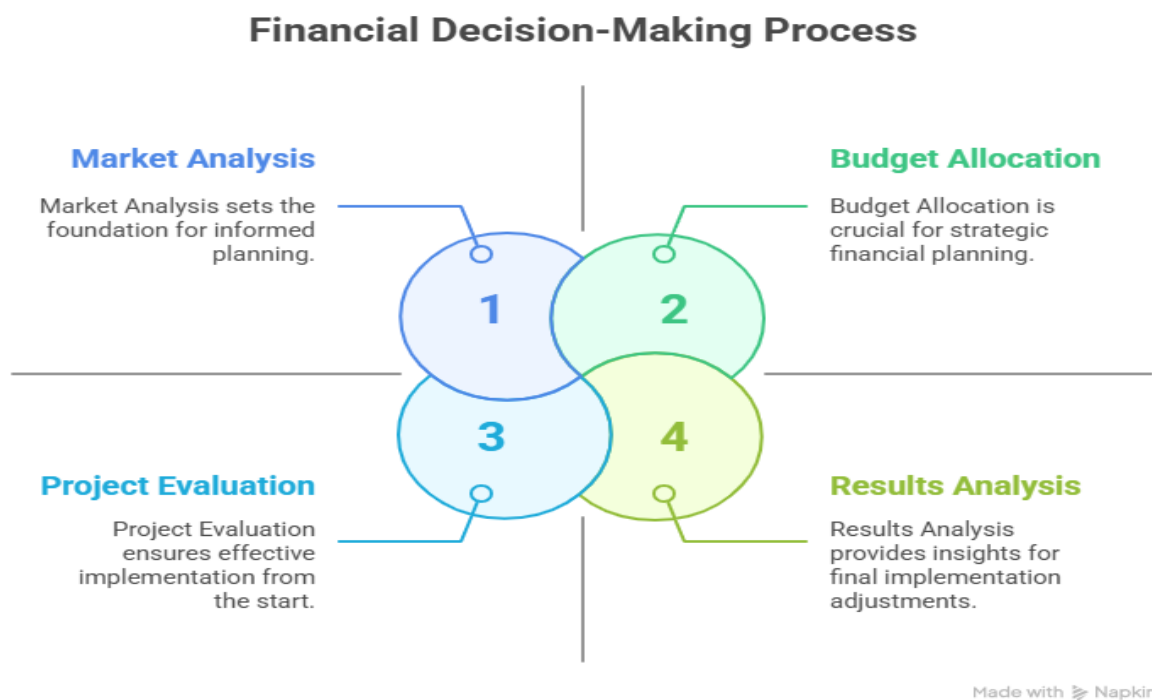
Budget Optimization Table:

Strategy	Projects Count	Average Investment	Risk Profile
Concentrated	3-4 large projects	15-25 million	Higher risk
Balanced	5-7 medium projects	8-15 million	Moderate risk
Diversified	8-10 smaller projects	5-10 million	Lower risk

Common Pitfalls:

- Exhausting entire budget without risk consideration
- Selecting projects based on investment size rather than returns
- Ignoring opportunity cost of excluded projects

5. Understanding Decision Linkages and System Dynamics



Explanation of Linkages:

- **Market → Risk:** Industry conditions directly influence project risk profiles
- **Risk → Discount Rate:** Higher risk requires higher discount rates
- **Evaluation → Portfolio:** Individual project analysis informs overall portfolio strategy
- **Budget → Selection:** Resource constraints drive final project choices
- **Metrics → Monitoring:** Financial projections establish performance benchmarks

Summary of Core Decision Areas & Key Parameters

Decision Area	Key Parameters	Interdependencies	Success Metrics
Market Analysis	Economic indicators, industry trends	Risk assessment, strategic planning	Forecast accuracy
Risk Assessment	Project complexity, market volatility	Discount rates, portfolio balance	Risk-adjusted returns
Discount Rate Setting	Cost of capital, risk premiums	NPV calculations, project rankings	Appropriate risk pricing
Project Evaluation	Cash flows, investment requirements	Budget allocation, synergies	Individual project performance
Portfolio Optimization	Budget constraints, diversification	Overall strategy, risk tolerance	Portfolio-level KPIs
Strategic Alignment	Business objectives, competitive position	All other decisions	Long-term value creation

6. Performance Analysis Framework

Key Performance Indicators (KPIs)

Primary KPIs:

1. Net Present Value (NPV) of Cash Stream

- Definition: Total present value of selected projects minus initial investments
- Calculation: $\Sigma(\text{Project NPV})$ for all selected projects
- Decision Area Impact:
 - Primary driver of project selection decisions
 - Influenced by discount rate accuracy
 - Reflects portfolio optimization effectiveness
- Interpretation: Higher NPV indicates superior value creation; optimal solution typically achieves $\text{NPV} > \text{INR } 200 \text{ million}$

2. Present Value (PV) of Cash Stream

- Definition: Total discounted future cash flows from selected projects
- Calculation: $\Sigma(\text{Discounted Cash Flows})$ across project duration
- Decision Area Impact:
 - Measures revenue generation potential
 - Affected by project mix and duration choices
 - Indicates growth trajectory strength
- Interpretation: Represents total expected returns; strong performance typically exceeds INR 350 million.

3. Average Discount Rate Across Projects (%)

- Definition: Weighted average of discount rates applied to selected projects
- Calculation: $\Sigma(\text{Project Investment} \times \text{Discount Rate}) / \text{Total Investment}$
- Decision Area Impact:
 - Reflects portfolio risk profile
 - Influenced by project category selection
 - Indicates risk management effectiveness
- Interpretation: Optimal range typically 8-12%; higher rates suggest excessive risk concentration

Summary of Key Performance Indicators

KPI	Target Range	Weight in Scoring	Primary Drivers
NPV of Cash Stream	>INR 200M	40%	Project selection, discount rates
PV of Cash Stream	>INR 350M	35%	Project cash flows, duration
Average Discount Rate	8-12%	25%	Risk assessment, project mix

7.Theoretical Connections by Simulation Section

Section	Simulation Area	Theory	Summary
Market Analysis	Industry Context	Market Efficiency Theory	Understanding how market conditions affect project valuations and risk assessments
Risk Assessment	Project Risk Profiling	Portfolio Theory (Markowitz)	Diversification benefits and systematic vs. unsystematic risk
Discount Rate Selection	Cost of Capital	CAPM and WACC Theory	Risk-adjusted required returns and capital structure impacts
Project Evaluation	Financial Metrics	DCF Valuation Theory	Time value of money and cash flow analysis principles
Portfolio Optimization	Resource Allocation	Modern Portfolio Theory	Efficient frontier and risk-return optimization
Budget Constraints	Capital Rationing	Capital Budgeting Theory	Optimal resource allocation under constraints
Performance Monitoring	Value Creation	Shareholder Value Theory	Measuring and maximizing economic value added

Critical Teaching Points

1. Risk-Return Relationship

- Higher risk projects require higher discount rates to compensate investors for increased uncertainty
- Portfolio diversification reduces overall risk through correlation effects between different project categories
- Market conditions influence all project valuations through changes in economic indicators and industry factors
- Risk assessment must consider both systematic (market-wide) and unsystematic (project-specific) factors

2. Capital Allocation Efficiency

- Limited resources require systematic prioritization frameworks to maximize value creation
- NPV rule provides optimal decision criterion

{What is the NPV Rule?}

The NPV (Net Present Value) rule is the fundamental decision criterion in capital budgeting that states: Accept all projects with positive NPV and reject those with negative NPV. When choosing among mutually exclusive projects or facing budget constraints, select the combination that maximizes total NPV. This rule ensures optimal allocation of capital by measuring the absolute value each project adds to the firm.}

- Budget constraints may force acceptance of suboptimal individual projects to achieve better overall portfolio performance
- Opportunity cost considerations become critical when capital is scarce

3. Time Value of Money Applications

- Present value calculations form the foundation of all investment decisions and project comparisons
- Discount rate selection critically impacts project rankings and can completely alter investment priorities
- Cash flow timing affects project attractiveness, with earlier cash flows being more valuable than later ones
- Understanding the relationship between risk, time, and return is essential for sound financial decision-making

4. Strategic vs. Financial Analysis

- Quantitative metrics must be balanced with strategic considerations such as market positioning and competitive advantage

- Portfolio effects may override individual project metrics when projects have synergistic relationships
- Long-term value creation requires integrated thinking that considers both financial returns and strategic positioning
- Real options value and strategic flexibility often justify investments that appear marginal on pure NPV analysis

8. Sample Optimal Solution

It is essential to acknowledge that numerous combinations can lead participants to achieve the desired outcomes, and the provided solution serves as a reference for faculty and is not intended to be shared directly with participants.

Please Note: The solution below is for the default Cesim Case
Project Portfolio Discount Rate

Project	Discount Rate
Global Outreach Initiative	11.03%
Regional Domination Campaign	9.45%
Digital Transformation Drive	11.45%
Emerging Markets Expansion	11.45%
Efficiency Overhaul Program	11.03%
Supply Chain Optimization	9.45%
Employee Training Enhancement	10.71%
Automated Claims Processing	11.24%
Customer Onboarding Redesign	10.71%
Regulatory Compliance Enhancement	11.45%
Data Security Strengthening	11.45%
Innovative Insurance Products	11.03%
Partnership for New Market Entry	11.24%
Customizable Policy Options	10.71%

Project Portfolio Selection

Project	Selection Status
Global Outreach Initiative	Not Selected
Regional Domination Campaign	Selected
Digital Transformation Drive	Not Selected
Emerging Markets Expansion	Not Selected
Efficiency Overhaul Program	Not Selected
Supply Chain Optimization	Selected
Employee Training Enhancement	Not Selected
Automated Claims Processing	Selected

Customer Onboarding Redesign	Selected
Regulatory Compliance Enhancement	Not Selected
Data Security Strengthening	Not Selected
Innovative Insurance Products	Selected
Partnership for New Market Entry	Not Selected
Customizable Policy Options	Not Selected

Portfolio Analysis Summary

- **Budget utilization:** 98.16% (INR 66.8 million out of INR 68 million)
- **Category allocation:** Market Expansion: 16.18%, Operational Efficiency: 13.24%, Process Improvement: 41.18%, Product Expansion: 27.94%
- **Risk profile:** Low (29.85%), Low-Moderate (13.43%), Moderate (28.36%), Moderate to High (28.36%)

Detailed Decision-Making Rationale

- **Strategic Portfolio Balance:** The chosen portfolio reflects a well-balanced investment across all four business areas, with a strong focus on Process Improvement and Product Expansion. This mix supports both operational excellence and future growth, aligning with the company's strategic maturity.
- **Risk-Adjusted Selection:** Projects were evaluated using differentiated discount rates based on risk profiles. Lower-risk, familiar initiatives received lower rates, while high-uncertainty projects were assigned higher rates, ensuring appropriate risk-return alignment.
- **Capital Efficiency:** Nearly full budget utilization demonstrates optimal capital deployment. Minimal unallocated funds reflect a disciplined application of NPV and capital rationing principles.
- **Category-Level Choices**
 1. Market Expansion: Focused on lower-risk regional growth over riskier global initiatives.
 2. Operational Efficiency: Selected projects with predictable returns and manageable complexity.
 3. Process Improvement: Prioritized automation and customer experience with strong ROI.
 4. Product Expansion: Chose scalable innovations over high-risk partnerships.
- **Risk Diversification:** The portfolio maintains a healthy mix of low to moderate-high risk projects, avoiding overexposure while enabling calculated growth.
- **Strategic Synergies:** Selected projects complement each other—operational upgrades support scalability, while product innovation pairs with improved customer experience and market reach.

9. Common Pitfalls and Teaching Interventions

Pitfall 1: Uniform Discount Rate Application

Symptoms:

- All projects assigned same discount rate
- Ignoring risk differentials between projects
- NPV calculations not reflecting true risk-adjusted returns

Intervention:

- Emphasize relationship between risk and required return
- Provide examples of how discount rates should vary by risk level
- Guide students through CAPM-based discount rate calculation
- Use sensitivity analysis to show impact of discount rate changes

Pitfall 2: NPV Maximization Without Risk Consideration

Symptoms:

- Selecting only highest NPV projects regardless of risk
- Ignoring portfolio diversification principles
- Concentrating investments in high-risk categories

Intervention:

- Introduce portfolio theory concepts
- Demonstrate risk-return trade-offs using historical examples
- Show how portfolio risk can be lower than individual project risks
- Encourage balanced allocation across project categories

Pitfall 3: Budget Exhaustion Strategy

Symptoms:

- Using entire INR 68 million budget without consideration
- Selecting projects to match exact budget amount
- Ignoring opportunity costs and risk management reserves

Intervention:

- Discuss importance of maintaining financial flexibility
- Explain opportunity cost concepts and future project possibilities
- Show examples of companies maintaining cash reserves
- Guide calculation of risk-adjusted budget allocations

Pitfall 4: Ignoring Market Context

Symptoms:

- Not incorporating industry indicators into analysis
- Using discount rates unrelated to market conditions
- Overlooking competitive landscape implications

Intervention:

- Review market analysis section systematically
- Connect macro-economic indicators to project valuations
- Discuss industry-specific risk factors
- Provide real-world examples of market impact on project success

Pitfall 5: Short-term Focus

Symptoms:

- Prioritizing projects with shortest payback periods
- Ignoring long-term strategic value creation
- Overemphasizing immediate cash flow generation

Intervention:

- Emphasize long-term value creation principles
- Show examples of strategic investments with longer payback periods
- Discuss the limitations of payback period as sole criterion
- Guide students through NPV vs. payback trade-off analysis

10. Instructor Tips and Insights

Exercises for Enhanced Learning

- Pre-simulation warm-up: Quick review of corporate finance fundamentals using real company examples
- Real-world connection: Use examples from recent corporate capital allocation decisions (e.g., Amazon's AWS investment, Apple's R&D spending)
- Scenario building: Create additional context about competitive pressures or regulatory changes
- Real-world Connection: Share current news about similar industry investments

Suggested Discussion Starters

1. How did the risk assessment philosophy influence the discount rate decisions, and what personal or professional biases can be recognized in the evaluation process?
2. Which project did the participants find hardest to evaluate financially, and how did they handle the uncertainty in the decision-making process?
3. How would the participants adapt their project selection strategy if the company's cost of capital suddenly increased by 3% due to rising interest rates?

4. What would be done differently if two of the selected projects require the same specialized team, creating a resource conflict?
5. How did you balance the tension between maximizing short-term NPV and building long-term competitive advantages through the project choices?
6. If the company's main competitor just announced a major digital transformation initiative, how would you reallocate your budget to respond strategically?
7. What assumptions about market conditions or project success rates are you most worried about, and how would you hedge against those risks?
8. How would the decision-making process change if you were making these choices for a startup versus an established company like this one?
9. If you had to choose between funding one high-impact INR 30 million project or three smaller INR 10 million projects, what factors would drive your decision?
10. How would you present your portfolio strategy to different stakeholders - shareholders focused on returns, employees concerned about job security, and customers expecting innovation?

Frequently Asked Questions (FAQs)

General Microsimulation Questions

Q: How is the final score calculated?

A: The final score is calculated based on three weighted Key Performance Indicators (KPIs): Net Present Value of Cash Stream (40%), Present Value of Cash Stream (35%), and Average Discount Rate Across Projects (25%). **Higher NPV and PV scores with an optimal discount rate range of 8-12% yield the best results.**

Q: How many rounds will we play in this Micro simulation?

A: The simulation typically consists of multiple rounds with maximum of 10 rounds, allowing to refine the strategy based on previous results. The instructor will specify the exact number of rounds for the course.

Q: Can we change my decisions after submission?

A: No, once decisions are submitted for a round, they cannot be altered. This design element emphasizes careful planning and thorough consideration before finalizing decisions.

Strategy Questions

Q: Is it better to choose all low-risk projects?

A: A balanced approach typically yields better risk-adjusted returns than extreme strategies.

Q: How important is diversification across project categories?

A: Very important. Concentration risk can significantly impact portfolio performance.

Technical Questions

Q: How do I determine appropriate discount rates?

A: Consider project risk level, market conditions, and company cost of capital as starting points.

Q: What if my NPV calculations don't match the system?

A: The system uses risk-adjusted cash flows that may differ from the stated cash flows based on risk multipliers.

Q: What if I can't spend the entire budget optimally?

A: Focus on maximizing NPV with available projects rather than forcing budget utilization.

Conceptual Questions

Q: Why might a lower NPV portfolio be better than a higher NPV one?

A: Risk considerations and portfolio balance may justify accepting lower absolute returns for better risk-adjusted performance.

Q: How do real companies make these decisions?

A: Companies use similar frameworks but with more detailed analysis, sensitivity testing, and strategic considerations.

Q: What role does intuition play versus quantitative analysis?

A: Both are important. Quantitative analysis provides the foundation, but business judgment is crucial for final decisions.

Conclusion

The Capital Budgeting Microsimulation provides a comprehensive, hands-on experience that transforms theoretical financial concepts into practical investment decision-making skills. Participants develop mastery of essential capital budgeting tools including NPV analysis, IRR calculations, and risk-adjusted discount rate estimation, while simultaneously building the strategic acumen necessary for effective portfolio optimization under realistic budget constraints.

Key Learning Outcomes:

1. **Technical Proficiency:** Mastery of capital budgeting fundamentals including NPV, IRR, payback period, and profitability index calculations in realistic business contexts
2. **Risk Assessment Skills:** Development of sophisticated risk evaluation capabilities through project categorization and discount rate estimation across diverse investment opportunities

3. **Strategic Resource Allocation:** Experience in optimizing investment portfolios while balancing competing priorities, budget limitations, and strategic objectives
4. **Critical Decision-Making:** Enhanced ability to navigate complex trade-offs between risk and return, short-term profitability and long-term value creation
5. **Portfolio Thinking:** Understanding of how individual project decisions aggregate into coherent corporate strategy and overall financial performance
6. **Professional Judgment:** Development of the analytical confidence and strategic mindset required for real-world financial leadership roles

By simulating real-world complexity and uncertainty, the experience builds both analytical confidence and strategic insight—preparing participants for impactful roles in corporate finance, investment planning, and strategic decision-making.